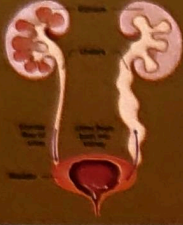


CHAPTER

10



Mutraghata

Learning Objectives

- ⇒ To learn the concept of Mutraghata
- ⇒ To know the Nidana Panchaka of Mutraghata mentioned in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To know the exact meaning of Mutra Shoshana and Mutra Pratihanana
- ⇒ To describe the different types of Mutraghata
- ⇒ To know the differences between Mutrakrichra and Mutraghata

Introduction

‘मूत्रकृच्छ्रे कृच्छत्वमतिशयितं, ईषद्विबन्धः; मूत्राघाते तु विबन्धो बलवान्, कृच्छत्वमल्पमिति

Madhu, Ma.Ni. – 31/1

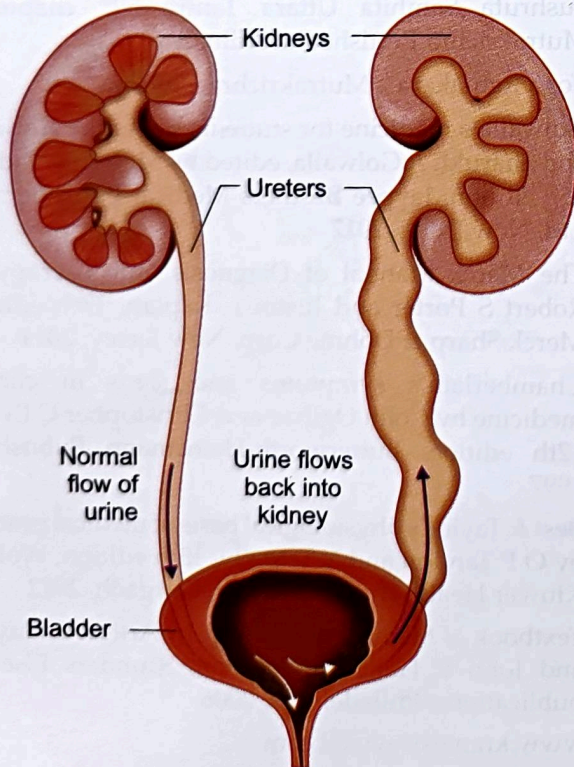


Figure 19 : Pratyatma Lakshana of Mutraghata

Mutraghata is a set of disease conditions where one can have less urine formation, but he will not have pain during micturition (*Mutrakrichra*) or the person may not have adequate urine output. *Charaka* says 13 types of *Mutraghata*. As per *Ayurveda* textbooks, the *Pramana* (amount) of *Mutra* formed in a day is 4 *Anjali Pramana*. As per the western physiology, the person will pass the urine once he gets the urge for micturition. Normally, an individual passes 200 – 250 ml of urine in one frequency of micturition. In *Mutraghata*, the person will have passage of less amount of urine¹. Normally, an individual will pass the urine with a normal range of 600 – 2500 ml in 24 hours with an average formation of urine is about 1200 to 1500 ml in 24 hours. When the individual passes < 500 ml in 24 hours, the condition is termed as *oliguria* and when the individual passes < 100 ml in 24 hours, then the condition is termed as *anuria*.

Nidana of Mutraghata

मूत्राघातो मूत्राबरोधः।

Ma.Ni. – 31/1

‘मूत्रकृच्छ्रे कृच्छत्वमतिशयितं, ईषद्विबन्धः; मूत्राघाते तु विबन्धो बलवान्, कृच्छत्वमल्पमिति।

Madhu, Ma.Ni. – 31/1

Nidana of *Mutraghata* as mentioned by *Madhavakara* are as follow:

- ❖ **Mutra Vegavidharana:** suppression of urge of micturition
- ❖ **Purisha Vegavidharana:** suppression of urge of

1 Less amount of urine will be passed in the conditions such as prostate enlargement, bilateral ureteric calculi, chronic renal failure, conditions which lead to renal hypoperfusion.

defaecation

❖ **Shukra Vegavidharana:** suppression of urge of ejaculation²

❖ **Rukshashana³:** excess consumption of dry substances

Purvarupa of Mutraghata: As per *Ayurveda* textbooks, no *Purvarupa* are described in the context of *Mutraghata*.

Pratyatma Lakshana of Mutraghata

मूत्राघातो मूत्रावरोधः।

Dalhana, Su.Ut. – 58/1,2

मूत्रकृच्छ्रे कृच्छ्रलमतिशयितं, ईषद्विबन्धः; मूत्राघाते तु विबन्धो बलवान्, कृच्छ्रलमल्पमिति।

Madhu, Ma.Ni. – 31/1

There are slight differences in the clinical understanding of *Mutrakrichra* and *Mutraghata*. As far as the cardinal feature of *Mutrakrichra*, there will be pain or discomfort during micturition where as in *Mutraghata*, the amount of urine passed in one frequency or in a day is reduced altogether. As per some other opinion, the term *Mutraghata* refers to complete absence of micturition.

Sankhya Samprapti of Mutraghata

Charaka and *Madhava* mentioned 13 types of *Mutraghata*, whereas *Sushruta* has mentioned 12 types of *Mutraghata*.

SN	Charaka	Sushruta	Madhava
1	Mutraukasada	Vatakundalika	Vatakundalika
2	Mutrathara	Asthila	Asthila
3	Mutrakrichra	Vatabasti	Vatabasti
4	Mutrotsanga	Mutrathita	Mutrathita
5	Mutra Sankshaya or Mutrakshaya	Mutrathara	Mutrathara
6	Mutrathita	Mutrotsanga	Mutrotsanga
7	Vatashthila	Mutrakshaya	Mutrakshaya

2 When an individual is suppressing these natural urges, there will be a vitiation of Apanavata initially. Vitiating Apanavata causes the reverse movement of Vata and thereby afflicts Samanavata which leads to the affliction of the Pranavata at the end. When there is affliction of Apanavata is observed, all functions related to Apanavata will be compromised – be it micturition or defecation etc. The vitiating Apanavata even may lead to the abnormal movement of Shukra in males and Artava in females also.

3 When an individual consumes more amount of Ruksha (dry) substances, that will lead to excess transport of fluid from intravascular compartment into alimentary canal compartment in the form of alimentary secretions. Thus, when an individual makes the habit of consuming more amount of Ruksha substances, that will lead to the gradual shrinkage of intravascular compartment, and when an individual develops renal hypoperfusion, then he will suffer from reduced urine formation.

4 बस्तिमेहनयोः शूलं मूर्धकृच्छ्रं शिरोरुजा। विनामो बंक्षणाहः स्याद्विहितं मूत्रनिग्रहे॥ 6॥

Ca.Su. – 7/6

The clinical features observed during the suppression of micturition are pricking type of pain in the urinary bladder and penis; difficulty in micturition; headache; forward bending of the body because of pain in lower abdomen and distension of the lower abdomen.

Micturition reflex is a completely autonomic spinal reflex, when an individual neglects the urge of micturition, then urination will not occur until the bladder fills still further and the micturition reflex becomes much stronger.

- As the person is suppressing the urge of micturition, this will lead to the constriction of urethral sphincter due to the activation of sympathetic nervous system.
- This will lead to the accumulation of urine in the urinary bladder and later there will be accumulation of urine in the ureter and renal pelvis.
- This will lead to the severe pain in the lower abdomen.
- This may be observed as a result of suppression of urge of micturition.

SN	Charaka	Sushruta	Madhava
8	Vatabasti	Mutragrathi	Mutragrathi
9	Ushnavata	Mutra Shukra	Mutrashukra
10	Vatakundalika	Ushnavata	Ushnavata
11	Mutragrathi	Pittaja Mutroukasada	Mutrasada
12	Vid Ghata	Kaphaja Mutroukasada	Vid Vighata
13	Bastikundala		Bastikundalika

Samprapti Ghataka of Mutraghata

Dosha : Vata Pradhana; Pitta and Kapha association

Dushya : Mutra

Srotas : Mutravaha Srotas

Srotodushti Prakara : Sanga and Vimargagamana

Udbhava Sthana : Pakvashaya

Sanchara Sthana : Sarvasharira

Vyakta Sthana : Basti and Upastha

Adhishtana : Basti

Vyadhyavastha : Asadhya

Vatakundalika

रौक्ष्याद्वेगविघाताद्वा वायुर्वस्तौ सवेदनः।

मूत्रमाविश्य चरति विगुणः कुण्डलीकृतः॥2॥

मूत्रमल्पाल्पमथवा सरुजं संप्रवर्तते।

वातकुण्डलिकां तां तु व्याधिं विद्यात्सुदारुणम्॥3॥ Ma.Ni. – 31/2,3

❖ *Vatakundalika* manifests in an individual due to suppression of urge of micturition⁴. This is leading to the increase of Ruksha (dry) quality in the body. This phenomenon leads to the vitiation of Apanavata.

❖ The person presents with pain in the Basti (lower abdomen) region.

- Upward movement of *Apanavata* and downward movement of *Mutra* during filling leads to the production of whirlpool inside the bladder.
- This phenomenon is called *Vatakundalika* which gives severe, intolerable and spasmodic pain the urinary bladder region.
- As *Mutra* moves in circular form inside the *Basti*, there is no excretion of urine.

Ashthila

आध्मापयन्बस्तिगुदं रुद्धवा वायुञ्च लोन्नताम्।
कुर्यात्तीवार्तिमष्टीलां मूत्र विष्मार्गरोधिनीम्॥४॥

Ma.Ni. - 31/4

- In the area⁵ between *Guda* and *Basti*, the vitiated *Apanavata* produces a hard stony mass.
- This hard mass will enlarge in size and in due course of time, this swelling obstructs the passage of urine, aching type of pain in genital organs and hypogastric region.
- This is called *Ashthila*, sometimes it is also called *Vatashthila*.

Vatabasti^{6,7}

वेगं विधारयेद्यस्तु मूत्रस्याकुशलो नरः।
निरुणद्धि मुखं तस्य बस्तर्बस्तिगतोऽनिलः॥५॥

मूत्रसङ्गो भवेत्तेन वस्तिकुक्षिनिपीडितः।

वातबस्तिः स विज्ञेयो व्याधिः कृच्छ्रप्रसाधनः॥६॥ Ma.Ni. - 31/5,6

- When the person suppresses the urge of micturition for long period of time, the bladder muscles lose their tonicity. The urethral opening gets closed due to *Apana Vata* vitiation.
- Due to the closure of outlet of urine transport, the suffering individual is not in a position to pass urine. Hence, urine gets obstructed and remains inside bladder.
- This condition produces pain and discomfort in severe form. This is called *Vatabasti*.

G
K

- In males, the organ situated between rectum and bladder is prostate. If there is any enlargement of prostate is observed, then the person may have oliguria or anuria.
 - This condition usually afflicts elderly individuals.
 - The person may complain of vague discomfort or fullness in the perineal or rectum area. The person may also complain of urgency (urgency for micturition, hesitancy (delay in initialising the micturition), intermittency (discontinuous flow of micturition). There may be also complaints of hematuria in some of the occasions.
 - Ideal examination performed in the assessment of prostate is per rectal examination. While doing this examination, we have to check the size of swelling, consistency, surface and mobility of the rectal mucosa over prostate should be analysed.
- This can be attributable to atonic bladder which results as a result of suppression of micturition for longer period of time.
- In this situation, the person will have the suppression of urge of micturition for longer period of time. In this instance, the person will have loss of tone of the urinary bladder for a transient period.
- In this condition, the person will have suppression of the urge of micturition, this will lead to the loss of tonicity of the urinary bladder. This will lead to stretching of the urinary bladder. When the next urge of micturition comes, it will come much stronger and more painful too. We can say it could be advanced stage of *Vatabasti*.
- In this situation, the person will have the habit of suppressing the urge of micturition, this will lead to the loss of tonicity of urinary bladder. Further, it will lead to distension of the urinary bladder and thence distension of ureters and later distension of the renal pelvis of both kidneys occur. This can be identified by swelling of suprapubic region. It may associate with severe abdominal pain, forward bending of the body and profuse sweating of the body.

Mutra Atita⁸

चिरं धारयतो मूत्रं त्वरया न प्रवर्तते।
मेहमानस्य मन्दं वा मूत्रातीतः स उच्यते॥७॥

Ma.Ni. - 31/7

- In this condition, the person who does the suppression of *Mutravega* i.e. urge of micturition for a certain period, later he attempts to void the urine, it may not be possible to excrete the urine.
- Therefore, the person strains or exerts more pressure to void the urine to initiate the act of micturition, but fails.
- When the person still exerts more pressure from abdomen, he may develop mild pain and may pass urine in little quantity after long continuous effort.
- Thus, *Mutratita* is a clinical condition where the patient pass the scanty, dribbling of urine after long continuous straining for micturition associated with pain.

Mutra Jathara⁹

सूत्रस्य वेगेऽभिहते तदुदावर्तहेतुकः।
अपानः कुपितो वायुरुदरं पूरयेद्भूशम्॥८॥
नाभेरधस्तादाध्मानं जनयेत्तीव्रवेदनम्।
तन्मूत्रजठरं विद्यादधोबस्ति निरोधनम्॥९॥

Ma.Ni. - 31/8,9

- Due to suppression of urge of micturition, the urine stays in *Basti* or urinary bladder. If the person still continues to suppress the urge of micturition the accumulation of urine in urinary bladder continues. Thus, the size of urinary bladder goes on increasing as the accumulation of urine continues and the patient still suppress the urge of micturition.
- The distension of urinary bladder reaches the level of umbilicus and distension will be visible externally. This is associated with severe abdominal pain. In this condition, the person will have obstruction to the passage of urine.

Mutra Utsanga¹⁰

वस्तौ वाऽप्यथवा नाले मणौ वा यस्य देहिनः।

मूत्रं प्रवृत्तं सज्जेत सरक्तं वा प्रवाहतः॥10॥

स्त्रवेच्छनैरल्पमल्पं सरुजं वाऽथ नीरुजम्।

विगुणानिलजो व्याधिः स मूत्रोत्सङ्गसंज्ञितः॥11॥ Ma.Ni. – 31/10,11

- In this clinical condition, due to the damage to the urinary bladder, urethra or chronic wound in the genitals the urethral channel becomes narrowed.
- Urine will pass with difficulty and dribbling of urine is seen. It may or may not be associated with pain. Sometimes, the urine is passed with blood.

Mutra Kshaya¹¹

रूक्षस्य क्लान्तदेहस्य बस्तिस्थौ पित्तमारुतौ।

मूत्रक्षयं सरुग्दाहं जनयेतां तदाह्वयम्॥12॥ Ma.Ni. – 31/12

- Due to emaciated body or increase of *Ruksha* (dry) quality of the body, there will be vitiation of *Pitta* and *Vata* in the *Basti* region.
- This is responsible for the decreased formation of urine and this is associated with burning sensation and severe pain.

Mutra Granthi¹²

अन्तर्बस्तिमुखे वृत्तः स्थिरोऽल्पः सहसा भवेत्।

अश्मरीतुल्यरुग्ग्रन्थिर्मूत्रग्रन्थिः स उच्यते॥13॥ Ma.Ni. – 31/13

- In this clinical condition, there will be small, spherical, stable mass or cyst. This is seen inside the urinary bladder, near the opening.
- This obstructs the passage of urine out of the urinary bladder.

Mutra Shukra¹³

मूत्रितस्य स्त्रियं यातो वायुना शुक्रमुद्धतम्।

स्थानाच्युतं मूत्रयतः प्राक् पश्चाद्वा प्रवर्तते॥14॥

भस्मोदकप्रतीकाशं मूत्रशुक्रं तदुच्यते। Ma.Ni. – 31/14,15

- When the person develops the urge of micturition, in stead of voiding the urine if the person withholds

and performs the sexual act, then there will be *Dushti* of *Shukravaha Srotas*.

- In this clinical condition, the urine is mixed with *Shukra* (semen) or the person may void *Shukra* followed with the flow of urine.
- The urine becomes thick and turbid and the colour resembles the colour of *Bhasmodaka* i.e. colour of water mixed with ash.

Ushnavata¹⁴

व्यायामाध्वातपैः पित्तं बस्तिं प्राप्यानिलान्वितम्॥15॥

बस्ति मेदं गुदं चैव प्रदहेत्स्त्रावयेदधः।

मूत्रं हारिद्रमथवा सरक्तं रक्तमेव वा॥16॥

कृच्छ्रात्पुनः पुनर्जन्तोरुष्णावातं ब्रुवन्ति तम्। Ma.Ni. – 31/15-17

- Due to the excessive physical exertion under hot sun, there will be increase of *Pitta* and manifestation of excessive sweating.
- Due to this mechanism, the person will pass little amount of concentrated urine.
- The person is presented with severe pain in the urinary bladder, genital, anus.
- The person may pass yellowish or reddish coloured urine or sometimes he may pass frank blood itself.

Mutra Sada¹⁵

पित्तं कफो द्वावपि वा संहन्येते ऽनिलेन चेत्॥17॥

कृच्छ्रान्मूत्रं तदा पीतं श्वेतं रक्तं धनं सृजेत्।

सदाहं रोचनाशङ्कुचूर्णवर्णं भवेत्तु तत्॥18॥

शुष्कं समस्तवर्णं वा मूत्रसादं वदन्ति तम्। Ma.Ni. – 31/17-19

- In this clinical condition, the person may pass yellowish, whitish or reddish discolouration of urine and density of urine is increased associated with burning sensation during micturition.
- When the urine is dried, the urine will have the appearance of powder of *Gorochana* (ox bile) in case of *Pitta* dominance and in case of *Kapha* dominance, the colour will be similar to the colour of *Shankha* (conch) when urine sample is made to dry.

- This may occur in the individual due to the damage to the urinary bladder or urethral stricture or wound to the genitals. As the diameter of the urethra is narrowed, this may lead to poor flow of urine to the exterior.
- In this clinical condition, the person may have renal hypoperfusion which in turn leads to the less formation of urine.
- This can be attributable to the bladder outlet obstruction which may occur as a result of either prostatomegaly or swelling occluding the urethra or urinary bladder, near the trigone area of bladder.
- In this condition, the presence of semen or sperms can be seen in the urine. This condition can be observed in the conditions of retrograde ejaculation.
- In this clinical condition, the person will have excess exposure to sunlight and there will be extreme sweating. This occurs due to the extreme heat produced in the body which leads to the shrinkage of the intravascular compartment. This will lead to the renal hypoperfusion which leads to the less formation of urine. In some of the occasions, the person may pass dark coloured urine and this can be attributable to poor glomerular filtrate..
- When the specific gravity of the urine is abnormally increased, there may be the presence of casts and crystals in the urine sample. This may be observed in case of dehydration or any condition where renal blood supply is compromised.

Vid Vigata¹⁶

रूक्षदुर्बलयोर्वातेनोदावृत्तं शकृद्यदा॥१॥

मूत्रस्वोतोऽनुपद्येत विदसंसृष्टं तदा नरः।

विद्वन्मूत्रयेत्कृच्छ्राद्विघातं विनिर्दिशेत्॥२०॥

Ma.Ni. - 31/19,20

- The person who is physically weak or who is having excess *Ruksha* (dry) quality in the body, this type of *Mutraghata* is seen.
- Initially, the person is presented with constipation which results in reverse movement of *Vata* i.e. *Udavarta*.
- Movement of faecal matter from *Purishavaha Srotas* to *Mutravaha Srotas*. This leads to the appearance of faecal matter in the urine, whenever the person gets the urge of micturition. The person may have faeculent smell in the urine due to the presence of faecal matter in the urine.

Bastikundala¹⁷

दृताध्वलङ्घनायासैरभिघातात्मपीडनात्।

स्वस्थानाद्वस्तिरुद्धतः स्थूलस्तिष्ठति गर्भवत्॥२१॥

शूलस्पन्दनदाहात बिन्दु बिन्दु स्रवत्यपि।

पीडितस्तु सृजेद्वारां संस्तम्भोद्वेष्टनार्तिमान्॥२२॥

वस्तिकुण्डलमाडुस्तं घोरं शस्त्रविषोपमम्।

पवनप्रबलं प्रायो दुर्निवारमबुद्धिभिः।

Ma.Ni. - 31/21-23

- मूत्राघातो मूत्रावरोधः।
- मत्रकृच्छ्रे कृच्छत्वमतिशयितं, ईषद्विबन्धः; मूत्राघाते तु विबन्धो बलवान्, कृच्छत्वमल्पमिति।
- *Mutraghata* is the spectrum of clinical conditions where we can observe formation of decreased amount of urine in a day. Normally, 200 to 250 ml of urine is passed in one frequency and 1200 to 1500 ml of urine is formed in 24 hours. If the amount is less than 500 ml in 24 hours, the clinical condition is termed as oliguria and if the amount of passed in 24 hours is less than 100 ml/day, then such clinical condition is termed as anuria. This clinical condition may manifest as a result of obstruction in the urethra, bladder neck, bilateral ureters or poor blood supply to the kidneys and clinical conditions afflicting the kidneys.
- In *Mutrakrichra*, the person will have difficulty in micturition and in case of *Mutraghata*, the amount of urine formed in a day is hampered significantly. This is one of the major difference between *Mutrakrichra* and *Mutraghata*.

Key points of the chapter

Dalhana, Su.Ut. - 58/1,2

Madhu, Ma.Ni. - 31/1

- 16 When an individual is having the habit of chronic constipation, that will lead to absorption of water in the faecal matter. Normally, faecal matter contains 75% of the water portion and 25% of the solid food residue.
 - When the person is suppressing the urge of defaecation intentionally for longer time, this will lead to the activation of sympathetic nervous system. When this is activated, that will lead to the constriction of the sphincter and this will also lead to the constriction of urethral sphincter.
 - This will lead to the hampered urine formation.
- 17 When an individual is exposed to such kind of measures which leads to the excess metabolism in the body such as excess walking or measures leading to the lightness of the body, will lead to shrinkage of intravascular compartment. This in turn may lead to the shrinkage of interstitial as well as intracellular compartments.
 - This will lead to the renal hypoperfusion and thus the person may have less urine formation.
 - As there is excess metabolism in the body, there is metabolic acidosis and there will be electrolyte imbalance due to sodium ions loss along with dehydration. This may lead to muscle cramps of the calf muscles or abdominal muscles too.
- 18 When there is bladder neck obstruction, we have to think two possibilities either it could be prostatomegaly or urethral stone impaction at the level of bladder neck. If this is not taken care of, then the person may have acute retention of urine.

Sadhyasadyata of Mutraghata

स्याद्वस्तौ कुण्डलीभूते तृणमोहः श्वास एव च॥२५॥

Ma.Ni. - 31/24

- If the *Kapha Dosha* is associated with bladder neck¹⁸, if *Pitta* vitiation is observed in a disease progression and if the urine is moving in the urinary bladder in circular fashion then such a disease is considered as *Asadhyata* (incurable) in nature.
- If there is no involvement of morbid *Vata*, then such a disease is considered as *Sadhyata* (curable) in nature.

- ⊗ No specific Nidana (aetiological factors) for the manifestation of Mutraghata is Ayurveda textbooks. But, Madhukosha commentary in the context of description of Mutraghata, he has described Mutra, Purisha, Shukra Vegadharana and Rukshapana are considered as Nidana of Mutraghata.
- ⊗ As far as Sankhya Samprapti of Mutraghata is concerned, Charaka and Madhavakara has mentioned 13 types of Mutraghata and Sushruta has mentioned only 12 types of Mutraghata. 13 types of Mutraghata are Vatakundalika, Ashthila, Vatabasti, Mutratita, Mutra Jathara, Mutrotsanga, Mutrakshaya, Mutragranthi, Mutrashukra, Ushnavata, Mutrasada, Vid Vighata, Bastikundala as per Madhavakara.
- ⊗ Mutraghata is a disease manifested due to the affliction of Apanavata with the involvement of Mutra, afflicting Mutravaha Srotas with Sanga and Vimargagamana as the type of Srotodushti involved in the progression of the disease. Pakvashaya is the Udbhava Sthana of the disease, Sanchara Sthana is Sarvasharira as the clinical features of Mutraghata can be observed all over the body and Vyakta Sthana of the disease is Basti and Upastha.

Self-Assessment: Questions for Reviews

Long essays

- ⊗ Long essay questions in this section are rare.

Short essays

- ⊗ Explain the Samprapti Ghataka of Mutraghata
- ⊗ Explain the types of Mutraghata
- ⊗ Explain Ushnavata

MCQs

1. Which of the following is included in the types of Mutraghata.

- a. Shitapitta; b. Amavata;
- c. Ushnavata; d. Amlapitta

Answer: (c)

2. Mutrakshaya, Mutrajathara and Mutraukasada are the types of

- a. Mutrakrichra; b. Mutraghata;
- c. Ashmari; d. Mutravridhi

Answer: (b)

3. Sankhya Samprapti of Mutraghata as per Sushruta is

- a. 13; b. 12;
- c. 8; d. 2

Answer: (b)

4. Which of the following is not the type of Mutraghata?

- a. Ashmarijanya; b. Mutrakshaya;
- c. Bastikundala; d. Ushnavata

Answer: (a)

5. Type of Srotodushti observed in Mutraghata is

- a. Sanga;
- b. Sanga and Vimargagamana;
- c. Atipravritti;
- d. Siragranthi

Answer: (b)

6. Vatakundalika is a Roga.

- a. Vataja; b. Mutraja;

- c. Amaja; d. Kshudra

Answer: (b)

7. Ushnavata is a type of

- a. Mutrakrichra; b. Mutraghata;
- c. Ashmari; d. Sharkara

Answer: (b)

8. Mutraghata is

- a. Retention of urine;
- b. Suppression of urine;
- c. Retention and suppression of urine;
- d. None of the above

Answer: (c)

9. Character of Mutra in Mutrashukra is

- a. Sarakta;
- b. Haridra Varna;
- c. Bhasmodaka Pratikasha;
- d. Nila Varna

Answer: (c)

10. Types of Mutraghata mentioned in Charaka Samhita in

- a. Sutra Sthana;
- b. Chikitsa Sthana;
- c. Siddhi Sthana;
- d. Nidana Sthana

Answer: (c)

11. In case of anuria, the amount of urine passed in 24 hours will be less than

- a. 100 ml;
- b. 500 ml;
- c. 1,000 ml;
- d. 3,000 ml

Answer: (a)

12. The Srotas afflicted in the disease Mutraghata is

- a. Mutravaha Srotas;
- b. Mamsavaha Srotas;

- c. Majjavaha Srotas;
- d. Raktavaha Srotas

Answer: (a)

13. Types of Mutraghata as per Madhavakara is

- a. 13; b. 12;
- c. 8; d. 3

Answer: (a)

14. The amount of urine passed in one frequency of micturition is

- a. 120 ml; b. 250 ml;
- c. 500 ml; d. 750 ml

Answer: (b)

15. Description of Mutraghata in Sushruta Samhita is available in

- a. Nidana Sthana; b. Chikitsa Sthana;
- c. Uttara Tantra; d. Kalpa Sthana

Answer: (c)

16. As per Madhavakara, Mutraghata is diagnosed by

- a. Ishat Mutra Vibandha;
- b. Adhika Mutra Vibandha;
- c. Adhika Mutra Krichratva;
- d. Ishat Mutra Krichratva

Answer: (b)

17. Which of the following symptom is not seen in the patients of prostatic enlargement?

- a. Hesitancy;
- b. Intermittency;
- c. Burning micturition;
- d. Continuous flow of urine

Answer: (d)

18. Which of the following term is used in case of the amount of urine < 500 ml is passed in 24 hours?

- a. Oliguria; b. Anuria;
- c. Nocturia; d. Polyuria

Answer: (a)

19. Which of the following feature will be absent in the patient suffering from acute retention of urine?

- a. Swelling of the suprapubic region;
- b. Forward bending of the body;
- c. Profuse sweating all over the body;
- d. Increased frequency of micturition

Answer: (d)

20. Which of the following region where swelling is observed in the case of acute retention of urine primarily?

- a. Epigastric region;
- b. Hypogastric region;
- c. Hypochondriac region;
- d. Umbilical region

Answer: (b)

Chapter resources

- Madhava Nidana, 31st chapter – Mutraghata Nidana Adhyaya
- Sushruta Samhita, Uttara Tantra – 58th chapter, Mutraghata Pratishedha Adhyaya
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- www.tamc.co.il
